Cover letter

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Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript “Unbounded strong-product ratios for the Grundy domination number” by Deep Bhattacharjee.

The paper proves that strong-product ratios for the Grundy domination number are unbounded, using an explicit two-field incidence construction. It also determines an all-parameter graph family and supplies a 114-vertex counterexample to multiplicativity. The general proofs are elementary and independent of finite computation.

A graph-theoretic counterexample and an unbounded family, with exact strong-product and domination results, match the journal’s graph and discrete-structure scope.

The prior public preprint is `https://hexagonmath.org/2610.00494v1`. The research manuscript/source archive is `https://doi.org/10.5281/zenodo.23607502`. Editable sources and verification records are supplied with the manuscript.

The Lean certificate verifies the original finite N=21 counterexample. The general unbounded-ratio theorem and all-parameter formula are written proofs, not a general Lean formalisation.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The submitting author has declared that all parts have been checked and understood by a human. The named human authors retain responsibility for the content.

Sincerely,

Deep Bhattacharjee
Corresponding author
itsdeep@live.com
ORCID: 0000-0003-0466-750X